INTERNAL MIGRATION NOTE: Google Colab to Kaggle Code Notebooks TO: Data Science Engineering & Analytics Team FROM: Ross Houslander / Lead Developer DATE: October 24, 2026 SUBJECT: Technical Evaluation and Migration Protocol: Transitioning Workflow from Google Colab to Kaggle Code Notebooks

1. Executive Summary This document outlines the workflow transition strategy for moving our data science notebook projects from Google Colab to Kaggle Code Notebooks. Because our recent Formula 1 dataset originated on Kaggle, migrating our runtime environment directly to Kaggle's native infrastructure streamlines data loading and eliminates external file-transfer bottlenecks. This note details what transfers natively, what requires re-engineering, and the primary operational risk associated with the move.
2. What Transfers Untouched (Zero-Modification Assets) The core intellectual property and logic of our notebooks require minimal refactoring: Jupyter Notebook Structure (.ipynb): Kaggle natively runs standard Jupyter-compatible notebook files. Markdown headers, text explanations, and cell order translate instantly. Core Python Logic & Syntax: Standard Python data manipulation libraries (pandas, numpy) and visualization libraries (matplotlib, seaborn) function identically without code modification.
3. What Needs Redoing (Environment & Infrastructure Redesign) Unlike local environments, moving between cloud platforms requires specific adjustments to state management, storage, and authentication: Account Setup & Permissions: Authentication shifts from a Google/Google Workspace account token to a verified Kaggle user profile equipped with an active API token (kaggle.json) if programmatic dataset fetching is required. File Storage & Dataset Linking: In Colab, we manually uploaded CSVs or mounted Google Drive. In Kaggle, datasets are attached directly as read-only inputs via the platform's input directory structure (typically mapped to /kaggle/input/). Code paths referencing local files must be updated from relative paths to /kaggle/input/[dataset-name]/. Package Installations & Runtime State: While standard libraries are pre-installed, any specialized non-standard packages must be re-initialized per session, or managed via Kaggle's environment configuration settings.
4. Critical Migration Risk: Ephemeral Storage and Session Timeouts The Risk: Kaggle Code Notebooks operate on strictly managed, ephemeral compute instances with hard execution limits (typically up to 9–12 hours for CPU/GPU sessions depending on the tier). If a notebook kernel idles out or exceeds runtime limits, local modifications made outside of designated persistent directories (/kaggle/working/) are permanently wiped. Mitigation: Ensure all generated outputs, cleaned dataframes, or exported models are explicitly saved to /kaggle/working/ and committed as notebook output versions or exported immediately prior to session termination.

Thank you for your time.